

Expander graphs: a primer

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Abstract

In this article we give a short combinatorial introduction to expander graphs, and discuss the importance of various measures connected to expanders such as the second largest eigenvalue of the adjacency matrix (λ_2), vertex expansion and spectral expansion. We also give a brief sketch of the inception of explicit construction of expanders which leads to certain expanders to be called as *Ramanujan graphs*.

1 Introduction

Expander graphs are sparse graphs that are D -regular, for a constant $D > 0$ having linear number of edges, however have good connectivity properties in terms of *edge expansion* or *vertex expansion*. The spectrum of a graph G is the set of eigenvalues of the adjacency matrix A of G . We refer to [Big93, Chapters 2 and 3] for an easy introduction to combinatorial properties of the spectrum of ordinary and regular graphs. From these properties it follows that the largest eigenvalue of a D -regular undirected graph G is D , and G is connected if and only if the multiplicity of D in the spectrum of G is 1. In addition, $D = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N \geq -D$, where λ_i is the i^{th} eigenvalue of the adjacency matrix A of a D -regular graph G , for $1 \leq i \leq N$. Also G is non-bipartite if and only if $\lambda_N > -D$ (also see [KS11, pp.15]).

1.1 Motivaton towards λ_2

In the eigenvalue spectrum of the adjacency matrix A of a D -regular graph G , the second largest eigenvalue (λ_2) of A is a key parameter that indicates the expansion properties of G . Now we motivate the importance of the second largest eigenvalue (λ_2) of the adjacency matrix A of a D -regular graph G .

For any $S \subseteq V(G)$, let $e(S)$ be the cardinality of the set of edges of the subgraph induced by S in G . Let us denote the set of edges connecting a vertex in S and a vertex in $\bar{S}$ by $E(S, \bar{S})$. Also let $e(S, \bar{S}) = |E(S, \bar{S})|$. A D -regular graph G is a δ -edge-expander (δ -expander for short) if for every set $S \subseteq V(G)$ of size at most $\frac{1}{2}|V(G)|$ there are at least $\delta D|S|$ edges connecting vertices in S and vertices in $\bar{S} = V(G) - S$, that is $e(S, \bar{S}) \geq \delta D|S|$.

Definition 1.1. Let G be a D -regular graph on N vertices and let $S \subseteq V(G)$ such that $|S| \leq N/2$. We define the edge-expansion ratio of G as

$$h(G) = \min_{\{S \mid |S| \leq \frac{N}{2}\}} \frac{e(S, \bar{S})}{|S|}$$

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Theorem 1.2. (Cheeger's inequality)[Che70] *Let G be a δ -expander and let A be the adjacency matrix of G with spectrum $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. Then*

$$\frac{D - \lambda_2}{2} \leq h(G) \leq \sqrt{2D(D - \lambda_2)}.$$

The above theorem due to J. Cheeger [Che70] signifies the importance of $(D - \lambda_2)$ known as the *spectral gap* of G . In particular, a δ -edge-expander G has an expansion ratio $h(G)$ bounded away from zero if and only if its spectral gap $(D - \lambda_2)$ is bounded away from zero.

Proof.

Proof of the left inequality: [ASS08, Theorem 1.1, pp. 324] Let A be the adjacency matrix of G and note that as A is symmetric we have $\lambda_2 = \max_{0 \neq x \perp 1^n} \langle xA, x \rangle / \langle x, x \rangle$. For a set $S \subseteq V(G)$ let x_S be the vector satisfying $x_i = 1$ when $i \in S$ and $x_i = 0$ otherwise, and note that $\langle x_S A, x_S \rangle = 2e(S)$ and $\langle x_S A, x_{\bar{S}} \rangle = e(S, \bar{S})$. Set $x = |\bar{S}| \cdot x_S - |S| \cdot x_{\bar{S}}$ and note that $x \perp 1^n$. Therefore,

$$\lambda_2(|S| + |\bar{S}|)|S||\bar{S}| = \lambda_2 \langle x, x \rangle \geq \langle xA, x \rangle = 2|S|^2 e(\bar{S}) + 2|\bar{S}|^2 e(S) - 2|S||\bar{S}| e(S, \bar{S}). \quad (1)$$

As G is D -regular, we have $e(S) = \frac{1}{2}(D|S| - e(S, \bar{S}))$ and $e(\bar{S}) = \frac{1}{2}(D|\bar{S}| - e(S, \bar{S}))$. Plugging this into (1), solving for $e(S, \bar{S})$ and using $|S| \leq n/2$, we complete the proof by inferring that

$$e(S, \bar{S}) \geq (D - \lambda_2)|S||\bar{S}|/n \geq \frac{1}{2}(D - \lambda_2)|S|.$$

Proof of the right inequality: [ASS08, Theorem 1.1, pp. 326] Let $Q = (DI - A)$ be Laplacian of G . Our goal is to prove that all but one of the eigenvalues of Q are at least $\frac{1}{2}\delta^2 D$. Let $z = (z_1, z_2, \dots, z_n)$ be an eigenvector of Q with the smallest nontrivial eigenvalue λ , where $V(G) = \{1, 2, \dots, n\}$. Recall that for every set U of at most half the vertices of G there are at least $c|U|$ edges between U and its complement, where $c = \delta D$ is some positive constant. Without loss of generality assume that $m \leq n/2$ of the entries of z are positive (otherwise, replace z by $-z$), and that $z_1 \geq z_2 \geq \dots \geq z_m \geq 0 \geq z_{m+1} \geq \dots \geq z_n$. Define $x_i = z_i$ for $i \leq m$, and $x_i = 0$ otherwise. Since $x_j = 0$ for all $j > n/2$,

$$\sum_{ij \in E(G)} |x_i^2 - x_j^2| = \sum_{ij \in E, i < j} (x_i^2 - x_j^2) \geq \sum_{i: i \leq n/2} (x_i^2 - x_{i+1}^2) ci = c \sum_{i=1}^n x_i^2. \quad (2)$$

Note that $(Qz)_i = \lambda z_i$ for all i and hence $\lambda = \frac{\sum_{i=1}^m (Qz)_i z_i}{\sum_{i=1}^m z_i^2}$. However,

$$\sum_{i=1}^m (Qz)_i z_i = \sum_{i=1}^m (dz_i^2 - \sum_{j, ij \in E} z_i z_j) = \sum_{i, j \leq m, ij \in E} (z_i - z_j)^2 + \sum_{i \leq m, j > m, ij \in E} z_i (z_i - z_j) \geq \sum_{ij \in E} (x_i - x_j)^2.$$

As $\sum_{i=1}^m z_i^2 = \sum_{i=1}^n x_i^2$ we conclude, using Cauchy-Schwartz inequality (twice) that

$$\lambda \geq \frac{\sum_{ij \in E} (x_i - x_j)^2}{\sum_{i=1}^n x_i^2} = \frac{\sum_{ij \in E} (x_i - x_j)^2 \sum_{ij \in E} (x_i + x_j)^2}{\sum_i x_i^2 \sum_{ij \in E} (x_i + x_j)^2} \geq \frac{(\sum_{ij \in E} |x_i^2 - x_j^2|)^2}{\sum_i x_i^2 2D \sum_i x_i^2} \geq \frac{c^2}{2D},$$

where the last inequality follows from (2). Therefore, $\lambda \geq \frac{c^2}{2D} = \frac{1}{2}\delta^2 D$. ■

1.2 Vertex expansion

Definition 1.3. Let $G = (V, E)$ be a simple undirected graph and let $N = |V|$. For any $S \subseteq V$, we denote the set of neighbors of vertices in S by $\Gamma(S)$. We say that G is an $(\alpha N, \Delta)$ -expander or that G has vertex expansion $(\alpha N, \Delta)$, if for every $S \subseteq V$ such that $|S| \leq \alpha N$ we have $|\Gamma(S)| \geq \Delta|S|$, where $0 \leq \alpha < 1$.

If G is a simple undirected bipartite graph with the partition of its vertex set into subsets L and R having N vertices each, then we say that G is an $(\alpha N, \Delta)$ -bipartite expander or that G has bipartite vertex expansion $(\alpha N, \Delta)$, if for every $S \subseteq L$ such that $|S| \leq \alpha N$ we have $|\Gamma(S)| \geq \Delta|S|$, where $0 \leq \alpha < 1$.

Theorem 1.4. Let $\mathcal{G}_{N,D}$ be the set of simple undirected bipartite graphs with bipartite sets L, R of size N each, and which is left regular having degree D for every vertex in L . Then for any D , there exists an $\alpha(D) > 0$ such that for all N

$$\Pr[G \text{ is an } (\alpha N, D-2)\text{-expander}] \geq 1/2,$$

where G is chosen independently and uniformly at random from $\mathcal{G}_{N,D}$.

Proof. To choose G in $\mathcal{G}_{N,D}$ uniformly at random, we choose for each vertex in L , D neighbors in R uniformly at random. For $K \leq \alpha N$, let us define

$$p_K = \Pr[\exists S \subseteq L : |S| = K, |\Gamma(S)| < (D-2)|S|].$$

So to prove that $\Pr[G \text{ is not an } (\alpha N, D-2)\text{-expander}] \geq 1/2$, it suffices to show that $\sum_K p_K \geq \frac{1}{2}$. Now assume that there is a set S of size K and $|\Gamma(S)| < (D-2)|S|$. Since the total number of neighbors of vertices in S , counted with multiplicity, is DK , there are at least $2K$ repeats among all the neighbors of vertices in S . We calculate the quantity,

$$\Pr[\text{at least } 2K \text{ repeats among all the neighbors of vertices in } S] \leq \binom{KD}{2K} \left(\frac{KD}{N}\right)^{2K}.$$

Here the binomial coefficient represents the number of ways of choosing $2K$ neighbors to be repeats from the (multi-)set of all neighbors of vertices in L , and the fraction $\frac{KD}{N}$ represents an upper bound on the probability that any given choice of a neighbor is a repeat. Since there are $\binom{N}{K}$ possibilities for S , we have,

$$\begin{aligned} p_K &\leq \binom{N}{K} \binom{KD}{2K} \left(\frac{KD}{N}\right)^{2K} \\ &\leq \left(\frac{Ne}{K}\right)^K \cdot \left(\frac{KD e}{2K}\right)^{2K} \cdot \left(\frac{KD}{N}\right)^{2K} \\ &\leq \left(\frac{cKD^4}{N}\right)^K, \end{aligned}$$

where $c = e^3$. By setting $\alpha = 1/(cD^4)$ and $K \leq \alpha N$, we know that $p_K \leq 4^{-K}$ and

$$\Pr[G \text{ is not an } (\alpha N, D-2)\text{-expander}] \leq \sum_{K=1}^{\alpha N} p_K \leq \sum_{K=1}^{\alpha N} 4^{-K} \leq 1/2.$$

■

Corollary 1.5. *Let G be a simple undirected graph chosen independently and uniformly at random from the set of all simple D -regular undirected graphs on N vertices. Then, for any D , there exists an $\alpha(D) > 0$ such that for all N ,*

$$\Pr[G \text{ is an } (\alpha N, D-2)\text{-expander}] \geq 1/2.$$

Proof. Given an D -regular undirected graph G on $\{1, \dots, N\}$ vertices, let us consider the D -regular undirected bipartite graph H on $2N$ vertices, labeled by $\{1, \dots, 2N\}$, obtained by having a partition of the vertex set of H into $L = \{1, \dots, N\}$ and $R = \{N+1, \dots, 2N\}$ containing N vertices each such that for every undirected edge (i, j) in G there exists an undirected edge $(i, j+N)$ in H . It is then easy to see that G is an $(\alpha N, D-2)$ -expander if and only if H is also an $(\alpha N, D-2)$ -expander. Our result now follows from Theorem 1.4. $\blacksquare$

1.3 Spectral expansion and vertex expansion

The normalized adjacency matrix M of a D -regular graph G is the matrix obtained by dividing the adjacency matrix A of G by D . Let G be a D -regular graph and let M be the normalized adjacency matrix of G . The largest eigenvalue of M is $\lambda_1 = 1$ with eigenvector $u = (1/N, \dots, 1/N)^T$. Then the second largest eigenvalue is given by,

$$\lambda_2 = \max_{\|x\|=1, x \perp u} \|Mx\|.$$

If π is a probability distribution on the vertices of G (represented as a vector), then we can write $\pi = u + \pi^\perp$, where $\pi^\perp \perp u$. Since G is D -regular and assuming π as the initial distribution,

$$M\pi - u = M(u + \pi^\perp) - u = Mu - u + M\pi^\perp = M\pi^\perp.$$

Thus,

$$\|M\pi - u\|^2 = \|M\pi^\perp\|^2 \leq \lambda_2^2 \|\pi^\perp\|^2 = \lambda_2^2 \|\pi - u\|^2.$$

Definition 1.6. *A graph G has spectral expansion λ if $\lambda_2(G) \leq \lambda$.*

Definition 1.7. *Given a probability distribution π , the collision probability of π is $\text{Coll}(\pi) = \|\pi\|^2 = \sum_x \pi_x^2$.*

Lemma 1.8. $\text{Coll}(\pi) = \|\pi - u\|^2 + 1/N$.

Proof. Write $\pi = u + \pi^\perp$. Then,

$$\|\pi\|^2 = \|u\|^2 + \|\pi^\perp\|^2 = 1/N + \|\pi - u\|^2.$$

$\blacksquare$

Note that $M\pi$ is also a probability distribution and using the lemma, we can compute the associated collision probability:

$$\text{Coll}(M\pi) - 1/N = \|M\pi - u\|^2 \leq \lambda^2 \|\pi - u\|^2 = \lambda^2 (\text{Coll}(\pi) - 1/N).$$

Given a probability distribution π , let the *support* of π be $\text{support}(\pi) = \{x | \pi_x \neq 0\}$.

Lemma 1.9. *Let π be a probability distribution. Then $\text{Coll}(\pi) \geq 1/|\text{support}(\pi)|$.*

Proof. Let $m = |\text{support}(\pi)|$. We claim that if $x_1 + x_2 + \cdots + x_m = x$, then $x_1^2 + x_2^2 + \cdots + x_m^2$ is minimized (with value x/m) when $x_1 = x_2 = \cdots = x_m = x/m$. This easily follows from the fact that $x^2 + y^2 \geq ((x+y)/2)^2 + ((x+y)/2)^2$. Thus $\text{Coll}(\pi) \geq 1/m$ and we are done. ■

Theorem 1.10. *Let G be a D -regular graph on N vertices. If G has spectral expansion λ , then for all $\alpha < 1$, G has vertex expansion $(\alpha N, \frac{1}{(1-\alpha)\lambda^2 + \alpha})$.*

Proof. Let $|S| \leq \alpha N$. Choose π to be the probability distribution that is uniform on S and 0 on the complement of S . Then,

$$\text{Coll}(\pi) = 1/|S| \quad \text{and} \quad \text{Coll}(M\pi) \geq 1/|\text{support}(M\pi)| = 1/|\Gamma(S)|.$$

Also,

$$1/|\Gamma(S)| - 1/N \leq \lambda^2(1/|S| - 1/N).$$

However $N \geq |S|/\alpha$, so solving the above inequality gives

$$|\Gamma(S)| \geq \frac{|S|}{(1-\alpha)\lambda^2 + \alpha}.$$

Thus G is an $(\alpha N, \frac{1}{(1-\alpha)\lambda^2 + \alpha})$ -expander. ■

Theorem 1.11. *Let G be a D -regular graph on N vertices which is a $(N/2, 1 + \Delta)$ -expander and let $\lambda_2(G)$ be the second largest eigenvalue of the normalized adjacency matrix M of G . If M has all non-negative eigenvalues, then G has spectral expansion λ , where $\lambda = 1 - \Delta^2/(d(8 + 4\Delta^2))$.*

Proof. Let x be an eigenvector with eigenvalue $\lambda_2(M)$. Since $x \perp u$, the vector x has both positive and negative entries. Let $V_+ = \{i : x_i > 0\}$ and $V_- = \{i : x_i \leq 0\}$. Without loss of generality $|V_+| \geq N/2$. Let $\bar{x}$ be the vector that agrees with x on V_+ and is 0 elsewhere.

Note that $\langle \bar{x}, \bar{x} \rangle = \langle x, \bar{x} \rangle$, so it can be shown that

$$\lambda_2(M) = \frac{\lambda_2 \langle x, \bar{x} \rangle}{\langle x, \bar{x} \rangle} = \frac{\lambda_2 \langle x, \bar{x} \rangle}{\langle \bar{x}, \bar{x} \rangle} = \frac{\langle Ax, \bar{x} \rangle}{\langle \bar{x}, \bar{x} \rangle}.$$

Also,

$$\begin{aligned} \lambda_2(M) \langle \bar{x}, \bar{x} \rangle &= \langle Ax, \bar{x} \rangle \\ &= \sum_{i,j} A_{ij} x_j \bar{x}_i \\ &= \|x\|^2 - \frac{1}{D} \left(D\|x\|^2 - \sum_{i \in V_+, \{i,j\} \in E} \bar{x}_i x_j \right) \\ &= \|x\|^2 - \frac{1}{D} \left(D\|x\|^2 - \sum_{i \in V_+, \{i,j\} \in E} \bar{x}_i \bar{x}_j - \sum_{i \in V_+, j \in V_-, \{i,j\} \in E} \bar{x}_i \bar{x}_j \right) \end{aligned}$$

$$\begin{aligned}
&= \|x\|^2 - \frac{1}{D} \left(D\|x\|^2 - \sum_{\{i,j\} \in E} \bar{x}_i \bar{x}_j \right) \\
&= \|x\|^2 - \frac{1}{D} \sum_{\{i,j\} \in E} (\bar{x}_i - \bar{x}_j)^2 \\
\lambda_2 &\leq 1 - \frac{\sum_{\{i,j\} \in E} (\bar{x}_i - \bar{x}_j)^2}{D \sum_{i \in V} \bar{x}_i^2}. \tag{3}
\end{aligned}$$

Build a new (weighted directed flow) graph H as follows. Let

$$V(H) = \{s\} \cup \{v_i : i \in V_+\} \cup \{w_j : j \in V_-\} \cup \{t\}.$$

For all $i \in V_+$, put the arcs (s, v_i) in H with capacity $1 + \Delta$. For each $i \in V_+$ and $j \in V_-$, where j is a neighbor of i in G , put the arcs (v_i, w_j) in H with capacity 1. Finally, for each $j \in V_-$, put the arcs (w_j, t) in H with capacity 1.

We claim that the minimum cut in this graph is $(1 + \Delta)|V_+|$. A cut of this size is given by the set of arcs $\{(s, v_i) : i \in V_+\}$. Given any other cut C , let $W = \{i \in V_+ : (s, v_i) \notin C\}$. It is easy to see that $|W| \leq N/2$, since otherwise C is not a minimum cut. For each $j \in \Gamma(W)$ in G , since j is a neighbor of i in G for some $i \in W$, it follows from the definition of H and the fact that C is a cut, that there must be an arc in C incident to w_j . However $|W| \leq N/2$ which implies $|\Gamma(W)| \geq (1 + \Delta)|W|$. So the capacity of C must be at least $(1 + \Delta)|V_+ - W| + |\Gamma(W)| \geq (1 + \Delta)|V_+|$ from which it follows that the minimum cut has capacity $(1 + \Delta)|V_+|$.

By the Max-flow min-cut Theorem [CLR22, Section 24.2, Theorem 24.6], there exists a flow on H of size $(1 + \Delta)|V_+|$. In particular, note that the flow through each vertex v_i must be $1 + \Delta$. Reading off the flow along arcs (v_i, w_j) , it follows that there is a function $F : V \times V \rightarrow \mathbb{R}$ satisfying the following conditions (here $\tilde{E}$ denotes the set of ordered pairs $\{i, j\}$ where $(i, j) \in E$, so that each edge in E is counted twice in $\tilde{E}$):

1. $0 \leq F(i, j) \leq 1$ for all $i, j \in V$.
2. $F(i, j) = 0$ if $i \notin V_+$ or $(i, j) \notin \tilde{E}$.
3. $\sum_{j: (i, j) \in \tilde{E}} F(i, j) = 1 + \alpha$ for each $i \in V_+$.
4. $\sum_{i: (i, j) \in \tilde{E}} F(i, j) \leq 1$ for each $j \in V$.

We need to calculate two bounds involving F in order to bound $\lambda_2(G)$. Keeping in mind that $2(a^2 + b^2) \geq (a + b)^2$ for all real numbers a and b , we find:

$$\begin{aligned}
\sum_{(i, j) \in \tilde{E}} F^2(i, j) (\bar{x}_i + \bar{x}_j)^2 &\leq 2 \sum_{(i, j) \in \tilde{E}} F^2(i, j) (\bar{x}_i^2 + \bar{x}_j^2) \\
&= 2 \sum_{(i, j) \in \tilde{E}} \bar{x}_i^2 \left(\sum_{(i, j) \in \tilde{E}} F^2(i, j) + \sum_{(i, j) \in \tilde{E}} F^2(j, i) \right)
\end{aligned}$$

$$\begin{aligned}
&\leq (4 + 2\Delta^2) \sum_{i \in V} \bar{x}_i^2. \\
\sum_{(i,j) \in \tilde{E}} F(i,j)(\bar{x}_i^2 - \bar{x}_j^2) &= \sum_{i \in V} \bar{x}_i^2 \left(\sum_{(i,j) \in E} F(i,j) - \sum_{(i,j) \in \tilde{E}} F(j,i) \right) \\
&\geq \Delta \sum_{i \in V} \bar{x}_i^2.
\end{aligned}$$

Note that in the third line, we used the fact that if $x_1 + \dots + x_n = 1 + \Delta$ and $0 \leq x_i \leq 1$ for all i , then $x_1^2 + \dots + x_n^2 \leq 1 + \Delta^2$. Multiplying equation (3) by

$$1 = \frac{\sum_{(i,j) \in \tilde{E}} F^2(i,j)(\bar{x}_i + \bar{x}_j)^2}{\sum_{(i,j) \in \tilde{E}} F^2(i,j)(\bar{x}_i + \bar{x}_j)^2}$$

and using Cauchy-Schwartz inequality, we get

$$\begin{aligned}
\lambda_2(G^2) &\leq 1 - \frac{\sum_{(i,j) \in E} (\bar{x}_i - \bar{x}_j)^2}{D \sum_{i \in V} \bar{x}_i^2} \\
&= 1 - \frac{\sum_{(i,j) \in E} (\bar{x}_i - \bar{x}_j)^2 \cdot \sum_{(i,j) \in \tilde{E}} F^2(i,j)(\bar{x}_i + \bar{x}_j)^2}{D \sum_{i \in V} \bar{x}_i^2 \cdot \sum_{(i,j) \in \tilde{E}} F^2(i,j)(\bar{x}_i + \bar{x}_j)^2} \\
&\leq 1 - \frac{\left(\sum_{(i,j) \in \tilde{E}} F(i,j)(\bar{x}_i^2 - \bar{x}_j^2) \right)^2}{2D(4 + 2\Delta^2) \cdot \left(\sum_{i \in V} \bar{x}_i^2 \right)^2} \\
&\leq 1 - \frac{\Delta^2}{D(8 + 4\Delta^2)}.
\end{aligned}$$

This completes the proof. ■

Definition 1.12. [BM08, pp. 82] *The k^{th} power of a simple graph $G = (V, E)$ is the graph G^k whose vertex set is V , two distinct vertices being adjacent in G^k if and only if their distance in G is at most k . The graph G^2 is referred to as the square of G , the graph G^3 as the cube of G .*

Let G be a undirected graph and let A be its adjacency matrix. Let us consider G as a directed graph G' in which every edge is replaced by a pair of directed edges in the opposite directions and every self-loop in G is replaced by a directed self-loop in G' . As a result, the number of directed closed walks of length k in G' is equal to the trace of A^k .

Now let G be an undirected graph, and without loss of generality assume that there exists a self-loop on every vertex of G . Let A be the adjacency matrix of G , then A^k is the adjacency matrix of G^k , for $k \in \mathbb{N}$. It is also easy to see that, for any square matrix A and $k \in \mathbb{Z}^+$, the eigenvalues of A^k are the k^{th} powers of the eigenvalues of A (see for example, [Str06, Section 5.2]). As a result, if G is a D -regular graph on N vertices and we assume without loss of generality that there exists a self-loop on every vertex of G , then G^k is a D^k -regular graph on N vertices and the eigenvalues of the adjacency matrix of G^k are the k^{th} powers of the eigenvalues of A , where $k \geq 1$ and A is the adjacency matrix of G .

We now move to the general case, when the eigenvalues of M need not all be non-negative.

Corollary 1.13. *Let G be a D -regular graph on N vertices which is a $(N/2, 1 + \Delta)$ -expander. Then G has spectral expansion λ , where $\lambda = \sqrt{1 - \Delta^2 / (D^2(8 + 4\Delta^2))}$.*

Proof. Consider the graph G^2 (recall Definition 1.12). If the normalized adjacency matrix of G is M , then the normalized adjacency matrix of G^2 is M^2 . Also G^2 is D^2 -regular and has all non-negative eigenvalues. Finally, G^2 is a $(N/2, 1 + \Delta)$ -expander, as follows: If S is a subset of the vertices of size at most $N/2$, then $|\Gamma(S)| \geq (1 + \Delta)|S|$. Choose a subset S' of $\Gamma(S)$ with $|S| \leq |S'| \leq N/2$. Then $|\Gamma(\Gamma(S))| \geq |\Gamma(S')| \geq (1 + \Delta)|S'| \geq (1 + \Delta)|S|$. But $\Gamma(\Gamma(S))$ is the neighborhood of S in G^2 , and S was an arbitrary subset of vertices of size at most $N/2$, so G^2 is a $(N/2, 1 + \Delta)$ -expander.

By Theorem 1.11, $\lambda_2(G^2) < 1 - \Delta^2/(D^2(8 + 4\Delta^2))$. Since the eigenvalues of G^2 are the squares of the eigenvalues of G , taking the square-root of the right-hand side proves the corollary. ■

1.4 Existence of Expander graphs

The existence of expander graphs was first proved by M. S. Pinsker [Pin73], [AS16, pp.152]. Standard probabilistic arguments (due to [Pin73]) show that almost every constant (≥ 3) degree graph is an expander (also see [ASS08, Theorem 2]). We recall (as stated in [JM87, pp. 343]) that the first explicit constructions of expanders are due to G. A. Margulis and also due to Gabber and Galil [GG81]. An elementary example of 3-regular expander graphs on p vertices, where p is a prime, is a Cayley graph defined as follows: let $G = (\mathbb{Z}_p, E)$, where for any vertex $x \in \mathbb{Z}_p$ edges $(x, x + 1)$, $(x, x - 1)$ and (x, x^{-1}) are in E . Here we assume that the multiplicative inverse of x is x^{-1} and that the multiplicative inverse of 0 is itself.

Several explicit constructions of expanders which are Cayley graphs are known. Constructions of families of bipartite expanders which are Cayley graphs having constant degree 5, 7, 8 shown in [GG81, JM87] are some of the earliest examples of expanders. As stated in [AS16, pp.153]: It is known (see [Nil91]) that the second largest eigenvalue of the adjacency matrix of any D -regular graph with diameter k is at least $2\sqrt{D-1}(1 - O(1/k))$. Therefore, in any infinite family of D -regular graphs, the second largest eigenvalue of the adjacency matrix is at least $2\sqrt{D-1}$ (see [KS11, Chapter 3], [LPS88, Proposition 4.2]). The importance of the number $2\sqrt{D-1}$ is due to N. Alon and R. Boppana [Alo86, pp. 95].

Definition 1.14. [LPS88, Definition 1.1] *Let $X_{N,D}$ be a D -regular graph on N vertices. A graph $X_{N,D}$ is a **Ramanujan graph** if $\lambda_2(X_{N,D}) \leq 2\sqrt{D-1}$.*

[LPS88, pp.262] have given explicit constructions of infinite families of D -regular Cayley graphs G_i , with the second largest eigenvalue $\lambda_2(G_i) \leq 2\sqrt{D-1}$ and hence these graphs are in fact Ramanujan graphs.

Definition 1.15. *Let G be a D -regular graph on N vertices and let A be the adjacency matrix of G . We say that G is a (N, D, λ) -expander if the second largest eigenvalue of A (denoted by λ_2) is at most λ .*

1.5 Significance of λ_2

The following theorem, known as the Expander Mixing Lemma, shows that every (N, D, λ) -expander graph is close to acting as a random D -regular graph on N vertices.

Theorem 1.16. [AC88] *Let $G = (V, E)$ be a D -regular, N -vertex graph with spectral expansion λ . Then $\forall S, T \subseteq V$, we have*

$$\left| |E(S, T)| - \frac{D|S| \cdot |T|}{N} \right| \leq \lambda \sqrt{|S||T|},$$

where $E(S, T)$ is the set of edges with one vertex in S and the other vertex in T .

Note that in the above statement, $|E(S, T)|$ is the number of edges with one vertex in S and the other vertex in T . Also in a random D -regular graph G , the expected number of edges between S and T in G is $\frac{D|S||T|}{N}$, since the $\Pr[\text{choosing a vertex randomly from any } S \subseteq V] = \frac{|S|}{N}$ and $|E| \leq ND$. As a result smaller λ implies that the expander graph G is more random.

Proof. Let A be the adjacency matrix of G . Since G is undirected, A is real and symmetric. As a result its eigenvalues are real and there exists a set of orthonormal eigenvectors $\{v_1, \dots, v_N\}$ of A which is a basis of $\mathbb{R}^N$. Let $\mathbf{1}_S$ and $\mathbf{1}_T$ be the characteristic vectors S and T . It is easy to see that

$$|E(S, T)| = \sum_{u \in S, v \in T} A_{uv} = \mathbf{1}_S^T A \mathbf{1}_T.$$

Expressing $\mathbf{1}_S$ and $\mathbf{1}_T$ as a linear combination of the orthonormal eigenvectors basis of $\mathbb{R}^N$, we have

$$\begin{aligned} |E(S, T)| &= \left(\sum_i \alpha_i v_i \right)^T A \left(\sum_j \beta_j v_j \right) \\ &= \sum_i \sum_j \beta_j \alpha_i v_i^T \lambda_j v_j \\ &= \sum_i \alpha_i \beta_i \lambda_i \\ &= \alpha_1 \beta_1 \lambda_1 + \sum_{i=2}^N \alpha_i \beta_i \lambda_i. \end{aligned}$$

We know that

$$\alpha_1 = \frac{|S|}{\sqrt{N}}, \beta_1 = \frac{|T|}{\sqrt{N}}, \lambda_1 = D.$$

Therefore,

$$\alpha_1 \beta_1 \lambda_1 = \frac{D}{N} |S| |T|$$

Then we get,

$$\begin{aligned} |E(S, T)| - \frac{D}{N} |S| |T| &= \sum_{i=2}^N \alpha_i \beta_i \lambda_i \\ \left| |E(S, T)| - \frac{D}{N} |S| |T| \right| &= \left| \sum_{i=2}^N \alpha_i \beta_i \lambda_i \right| \end{aligned}$$

We can bound the right side as,

$$\begin{aligned} \left| \sum_{i=2}^N \alpha_i \beta_i \lambda_i \right| &\leq \lambda \cdot \sum_{i=2}^N |\alpha_i \beta_i| \\ &\leq \lambda \left(\sum_{i=2}^N |\alpha_i|^2 \right)^{\frac{1}{2}} \left(\sum_{i=2}^N |\beta_i|^2 \right)^{\frac{1}{2}} \\ &\leq \lambda \cdot \|\mathbf{1}_S\|_2 \|\mathbf{1}_T\|_2 \\ &= \lambda \sqrt{|S| |T|} \end{aligned}$$

As a result,

$$\left| |E(S, T)| - \frac{D|S| \cdot |T|}{N} \right| = \left| \sum_{i=2}^N \alpha_i \beta_i \lambda_i \right|$$

which implies,

$$\left| |E(S, T)| - \frac{D|S| \cdot |T|}{N} \right| \leq \lambda \sqrt{|S||T|}.$$

■

The result proved in [BL06, Lemma 3.3 & Corollary 5.1] is the converse of the Expander Mixing Lemma shown above.

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